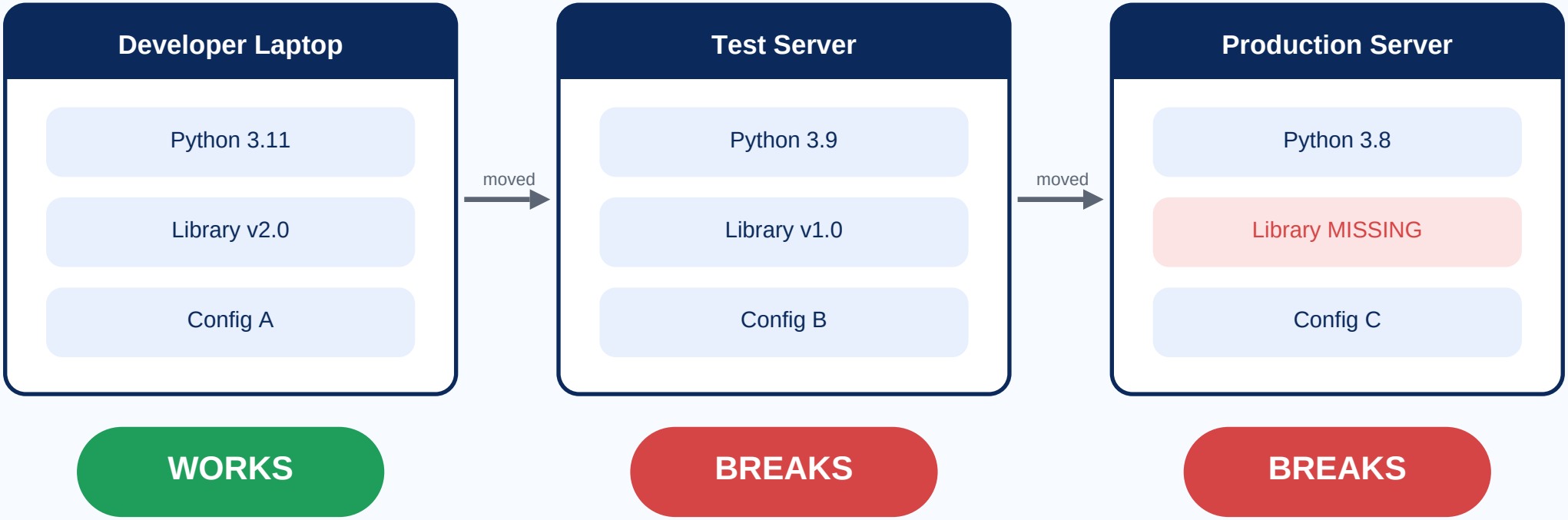
1. The Problem: "It works on my machine"

Same app, three different computers, three different results

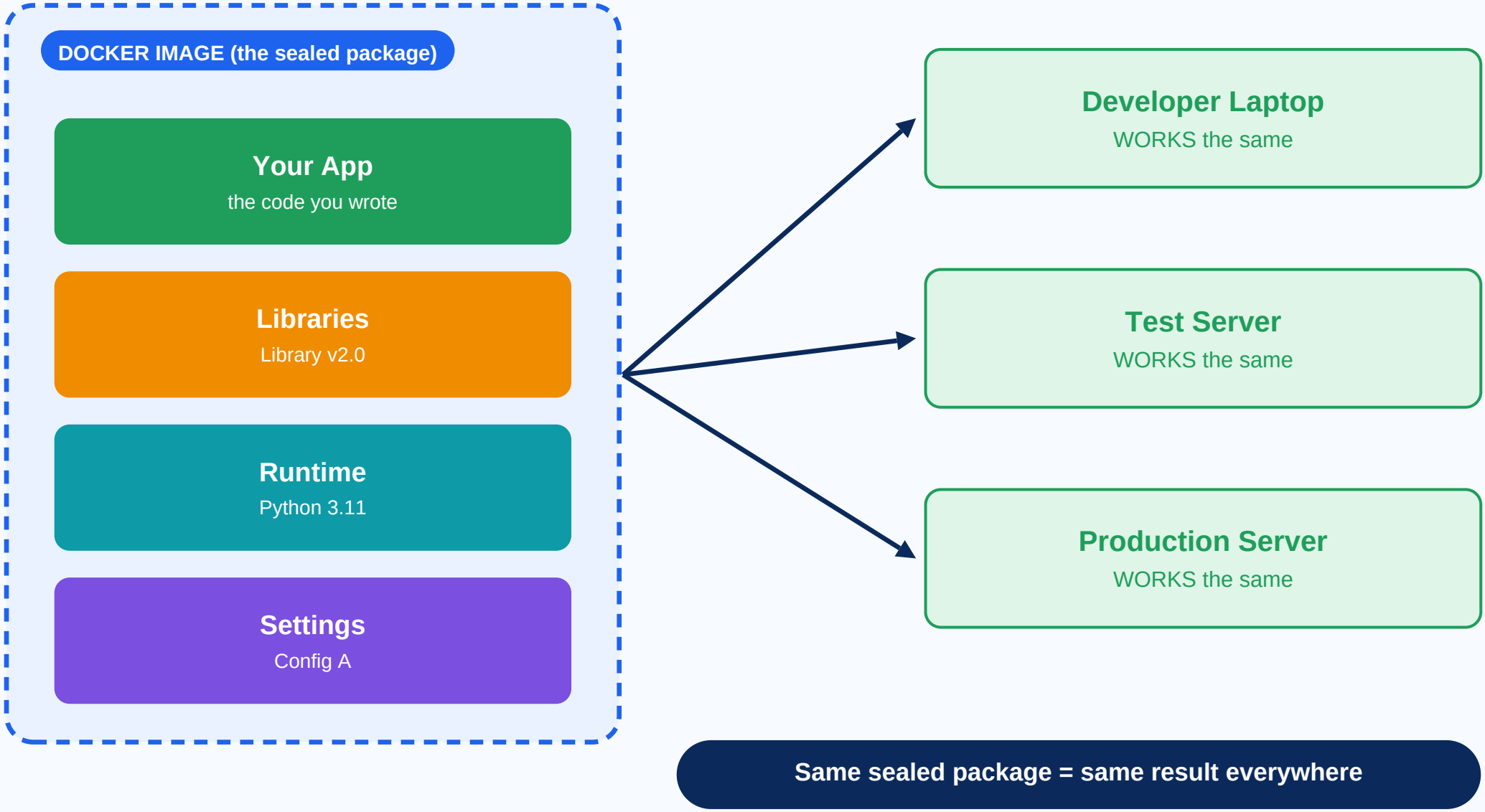


Every computer is a little bit different

Different language versions, missing libraries, different settings. The same app on a different computer gives surprise failures, and the developer says: "But it works on my machine!"

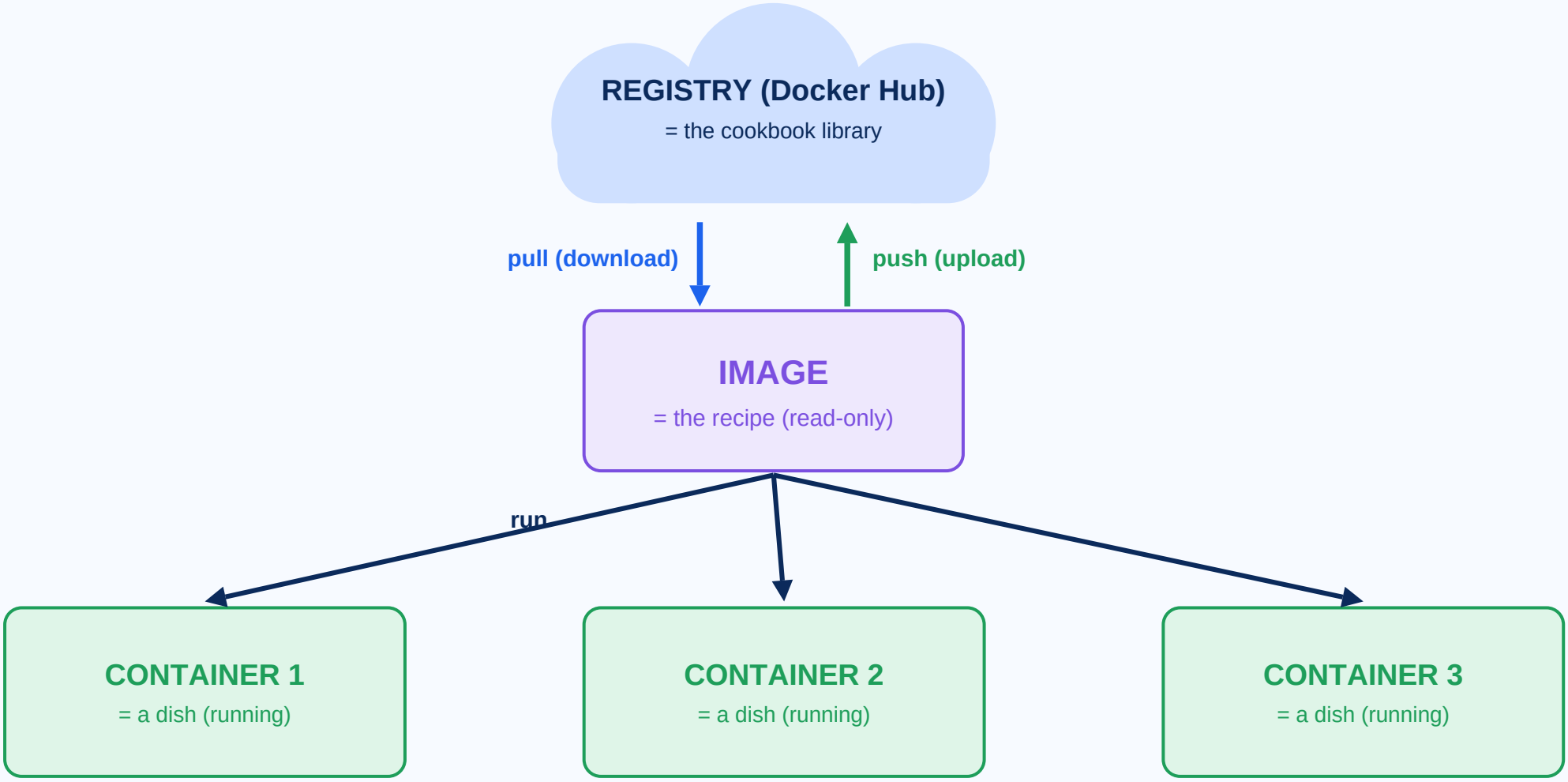
2. The Solution: Pack Everything Together

Docker bundles the app and all it needs into one sealed package



3. Image, Container and Registry

The three words that explain almost everything in Docker



One image can create MANY containers, just like one recipe can make many dishes.

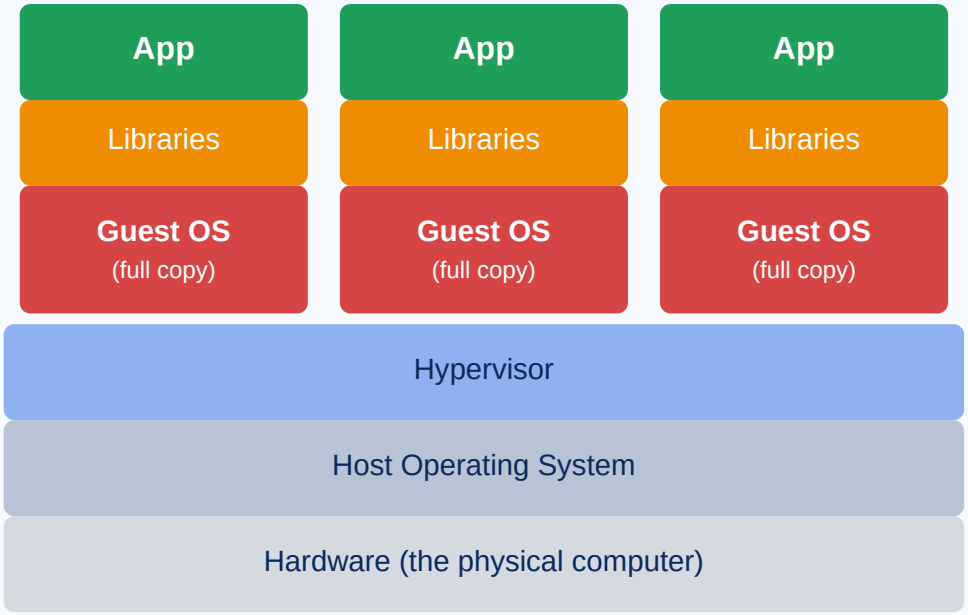
4. Containers vs Virtual Machines

Apartments in one building vs separate houses

VIRTUAL MACHINES

Each app carries a whole operating system

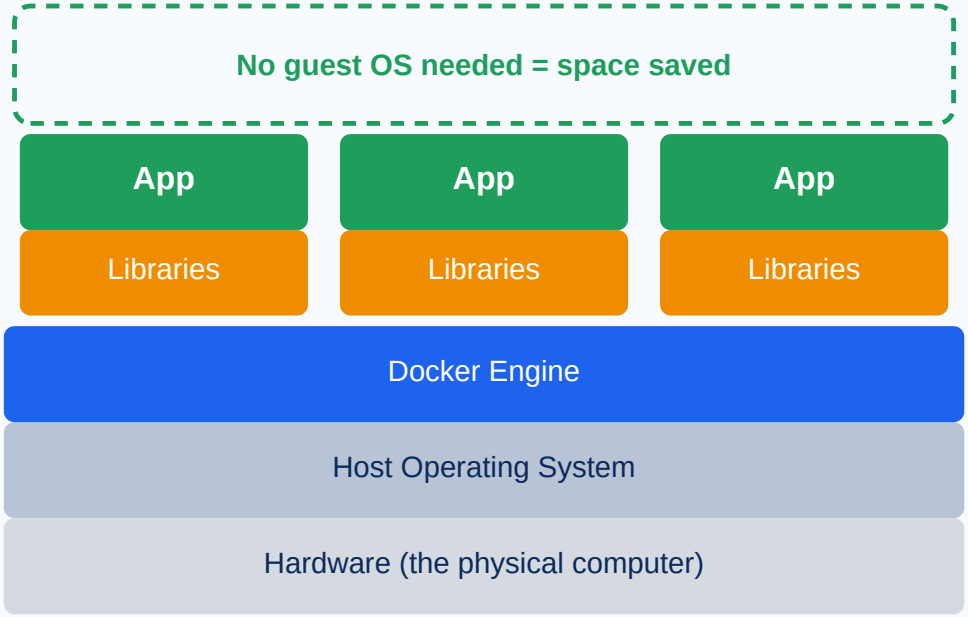
Size: gigabytes | Start: minutes | Run: a few at once



CONTAINERS

All apps share the host's operating system

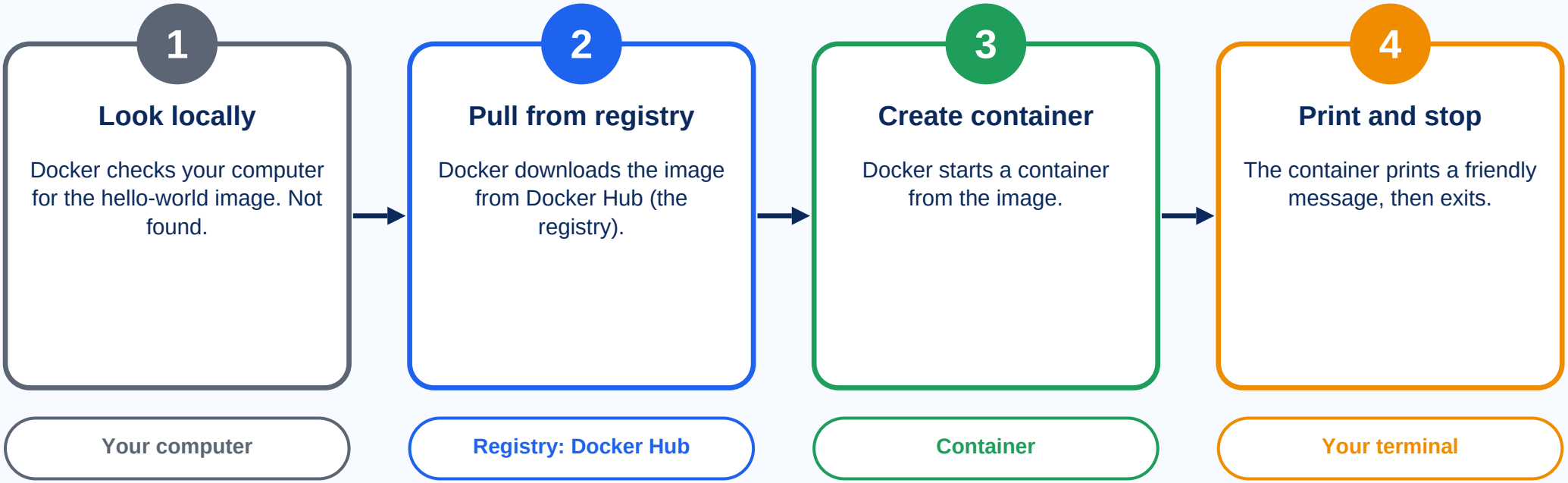
Size: megabytes | Start: seconds | Run: dozens



5. What Happens When You Run: docker run hello-world

One command, four steps. You will use an image, a registry and a container

```
$ docker run hello-world
```



In just one command you used all three ideas

Registry (where it came from) + Image (the blueprint) + Container (the running copy)

6. Before and After Docker

What changes for a DevOps team

BEFORE
Every new server

Developer builds the app → Ops installs wrong Python version → A library is missing → Hours of debugging → Repeat for every server

AFTER

Developer builds an image → Ops runs one command → It works. Done!

Why DevOps teams love Docker

Consistency	Speed	Efficiency	Isolation	Easy deploys	CI/CD ready
Same package everywhere	Starts in seconds	Many per machine, lower cost	One crash stays contained	Replace one container	Easy to automate